

EMPLOYEE WORKFORCE INTELLIGENCE SYSTEM

PROJECT OVERVIEW

The Employee Workforce Intelligence System is an HR analytics project designed to transform workforce data into clear, actionable insights. The system analyzes employee headcount, workforce status, payroll, tenure, departmental distribution, and attrition to support data-driven workforce management and HR decision-making.

BUSINESS PROBLEM

HR teams need a clear view of workforce size, employee status, departmental staffing, payroll, tenure, and attrition. Without structured workforce analytics, identifying staffing gaps and retention concerns can become difficult. This project addresses that need by organizing employee data into a simple workforce intelligence system.

DATASET & DATA STRUCTURE

The dataset contains 50 synthetic employee records created for portfolio-based HR analytics. Each record includes employee demographics, department, designation, joining date, employment type, employment status, salary, work shift, manager, and tenure. The structured dataset enables workforce-level and department-level analysis.

KEY ANALYTICS AREAS

The system focuses on key HR analytics areas including workforce headcount, employee status, departmental distribution, payroll analysis, average salary, employee tenure, and attrition. These metrics provide a structured view of workforce composition and help identify important workforce trends.

DASHBOARD & DECISION SUPPORT

The HR Dashboard presents key workforce KPIs and department-level analytics in a visual format. It enables quick monitoring of workforce size, active and resigned employees, payroll, salary, tenure, and attrition. The dashboard is designed to support management review and data-driven HR decision-making.

KEY WORKFORCE INSIGHTS

The analysis shows a workforce of 50 employees, with 44 currently active and an overall attrition rate of 10%. IT has the highest department-level attrition rate at 75%, while Sales & Marketing records 20%. These insights can help HR identify areas requiring closer workforce monitoring and retention-focused analysis.

TOOLS & TECHNOLOGIES

The project was developed using spreadsheet-based data management and HR analytics techniques. The primary tools include Excel/Google Sheets for data organization, calculations, dashboard creation, and visualization. The project also applies fundamental HR analytics concepts such as KPI tracking, workforce segmentation, payroll analysis, tenure analysis, and attrition analysis.

PROJECT LIMITATIONS

This project uses synthetic workforce data created for portfolio and learning purposes. The analysis is intended to demonstrate HR analytics concepts and does not represent the actual workforce, policies, or performance of any organization. The system can be further enhanced with real organizational data, advanced statistical analysis, SQL, Power BI, and predictive analytics.

PROJECT STATUS

The Employee Workforce Intelligence System has been completed as a structured HR analytics portfolio project. It includes an employee master dataset, HR calculations, department-level analysis, an interactive workforce dashboard, and project documentation. The project demonstrates an end-to-end approach to transforming workforce data into actionable HR insights.